Amazones



Présenté par
Nicolas Caminade, Tanguy Decabrat, Anya Bazizi, Younes Ouaammou, Thomas-Alexandre Moreau

Sommaire

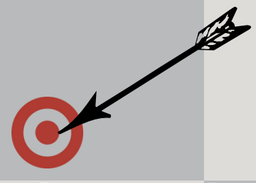


Le jeu

Règles

Histoire

Complexité

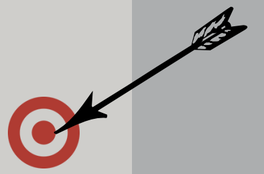


Structures, algorithmes et architecture interne

Logique de jeu

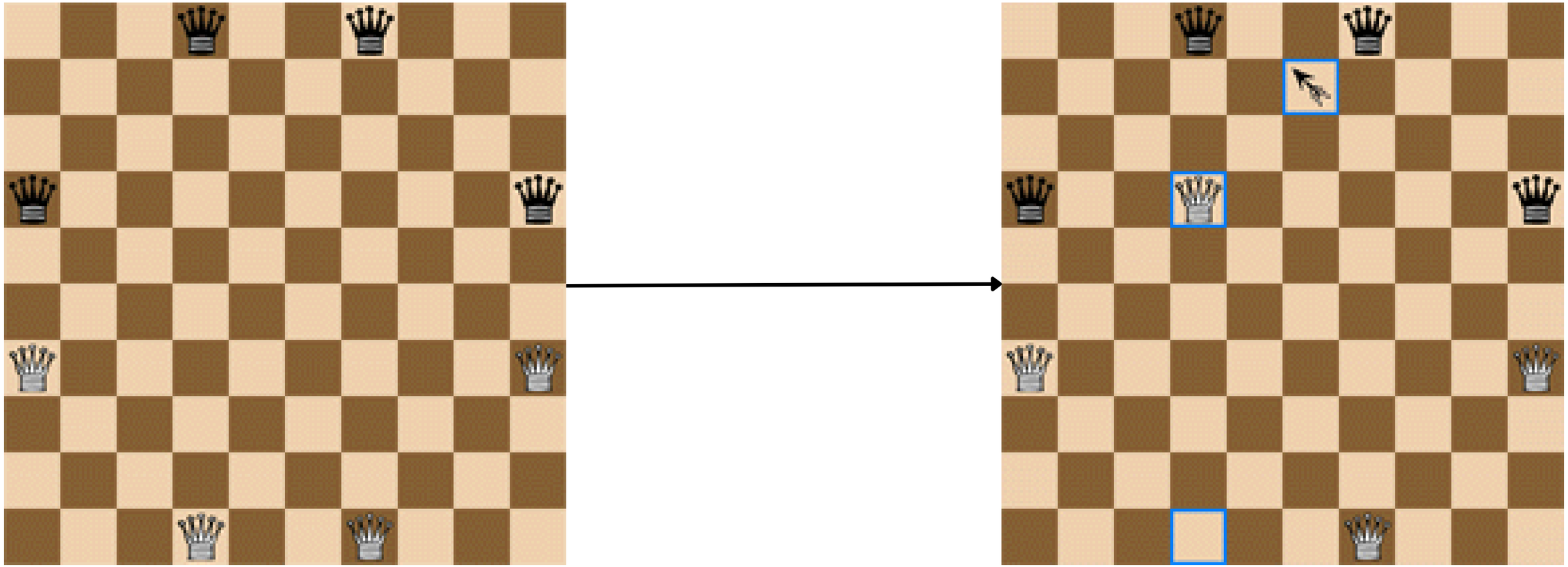
Réseau

IA



Architecture globale

Règles



Histoire du jeu

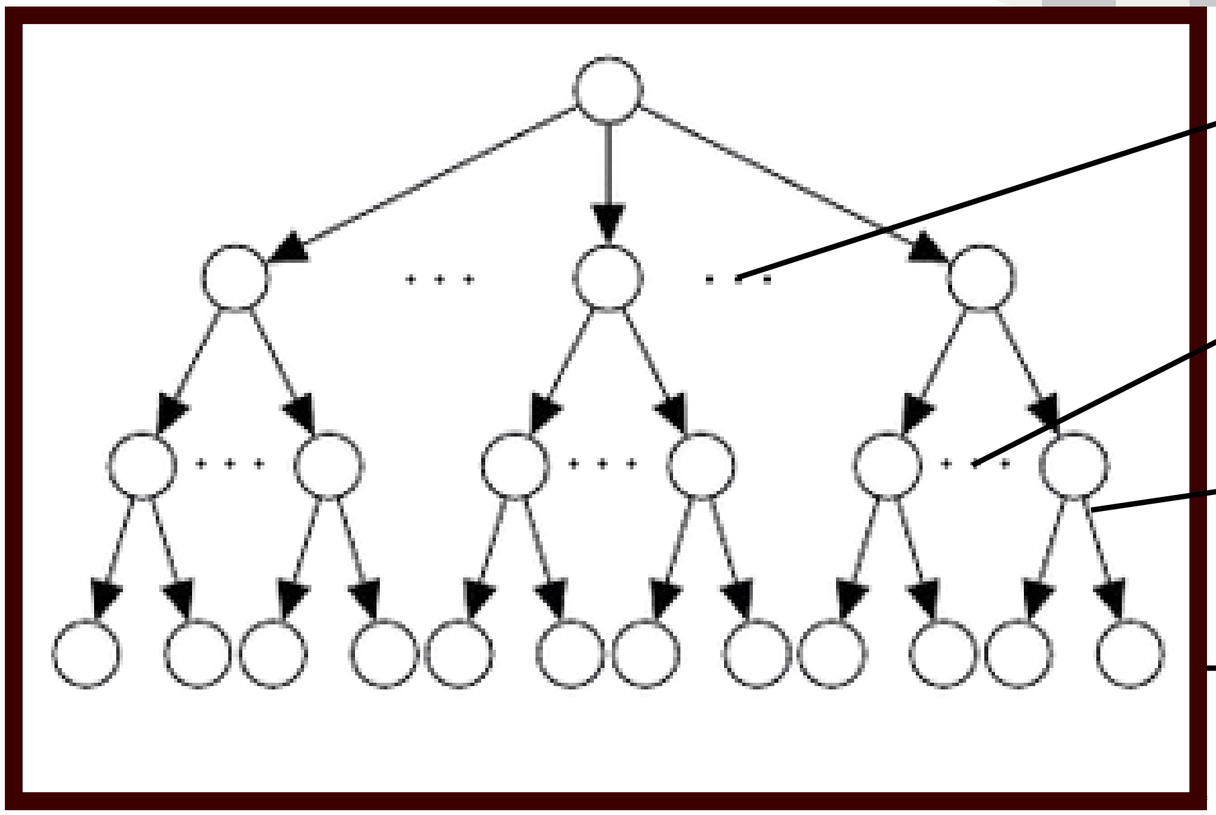


La Preuve de Complexité (2005)

La Révolution de l'IA et le MCTS (2000-2015)

L'ère du Deep Learning (Depuis 2017)

Complexité



Informations tirées de :
Searching for Solutions in Games and Artificial Intelligence
&
Combinatorics of Go

NB coups départ : ~2176

NB coups moyen: ~374/~299

Nb. Tours Moyen : ~84

Nb. parties possibles : ~10^212

unique + transpose

Nb. plateaux possibles : ~10^40

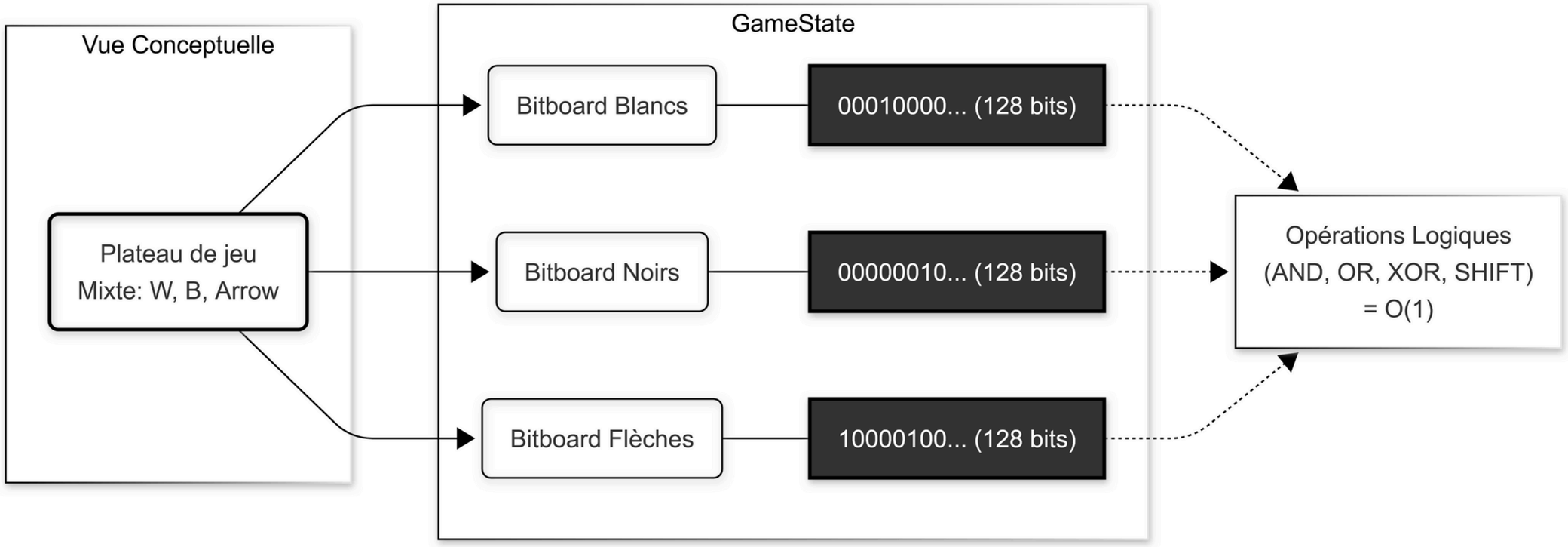
Et par rapport à d'autres jeux ?

	Échecs	Amazones	Go
Facteur Branch. Initial	~ 20	~ 2176	~ 361
Facteur Branch. Moyen	~ 35	~ 374 joueur de départ ~ 299 deuxième joueur	~ 250
Nombre de tours moyen	~ 80	~ 84	~ 150
Cmplx. Arbre de jeu	$\sim 10^{123}$	$\sim 1.09 * 10^{212}$	$\sim 10^{360}$
Cmplx. Espaces d'états	de $\sim 3.7 * 10^{43}$ à $\sim 10^{52}$	$\sim 6.45 * 10^{40}$	$\sim 10^{170}$

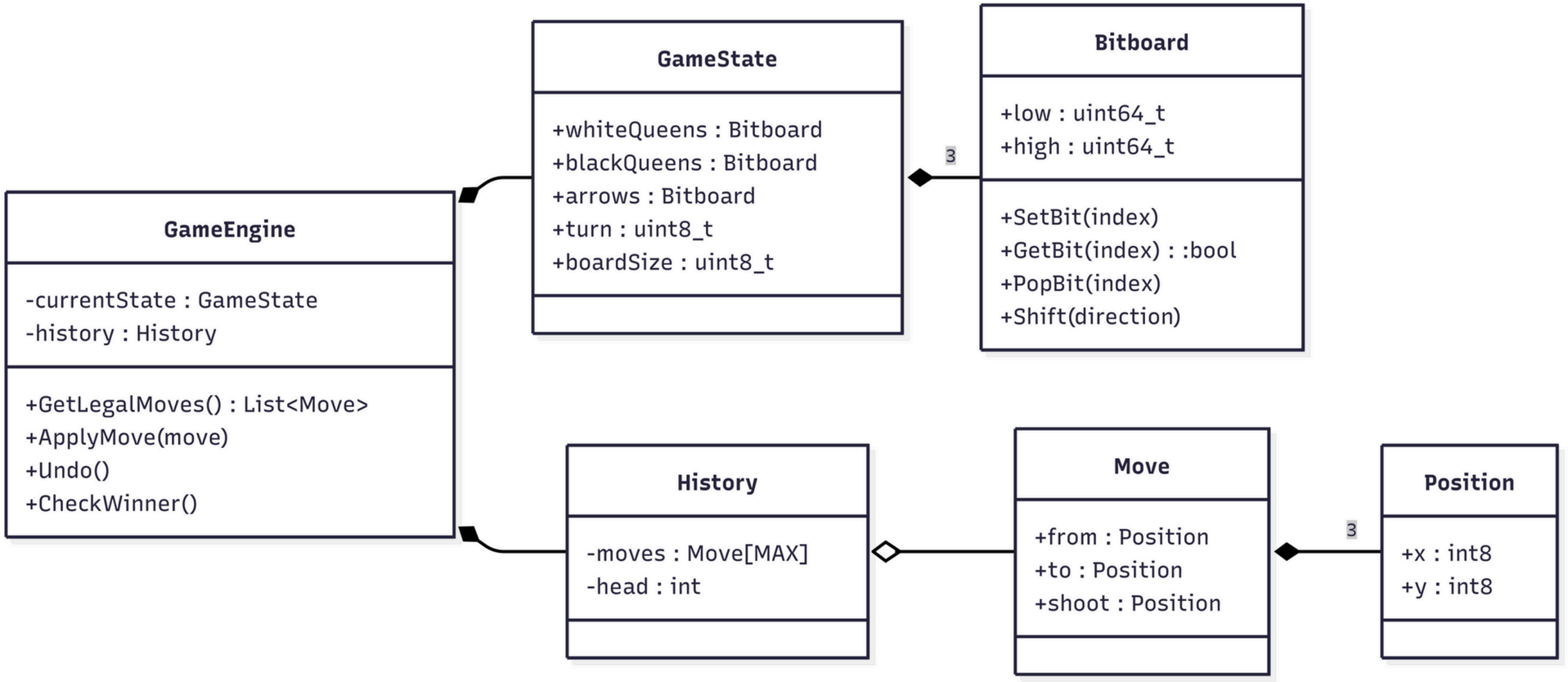
Informations tirées de :
Searching for Solutions in Games and Artificial Intelligence
&
Combinatorics of Go

Algorithmes et Structures de Données

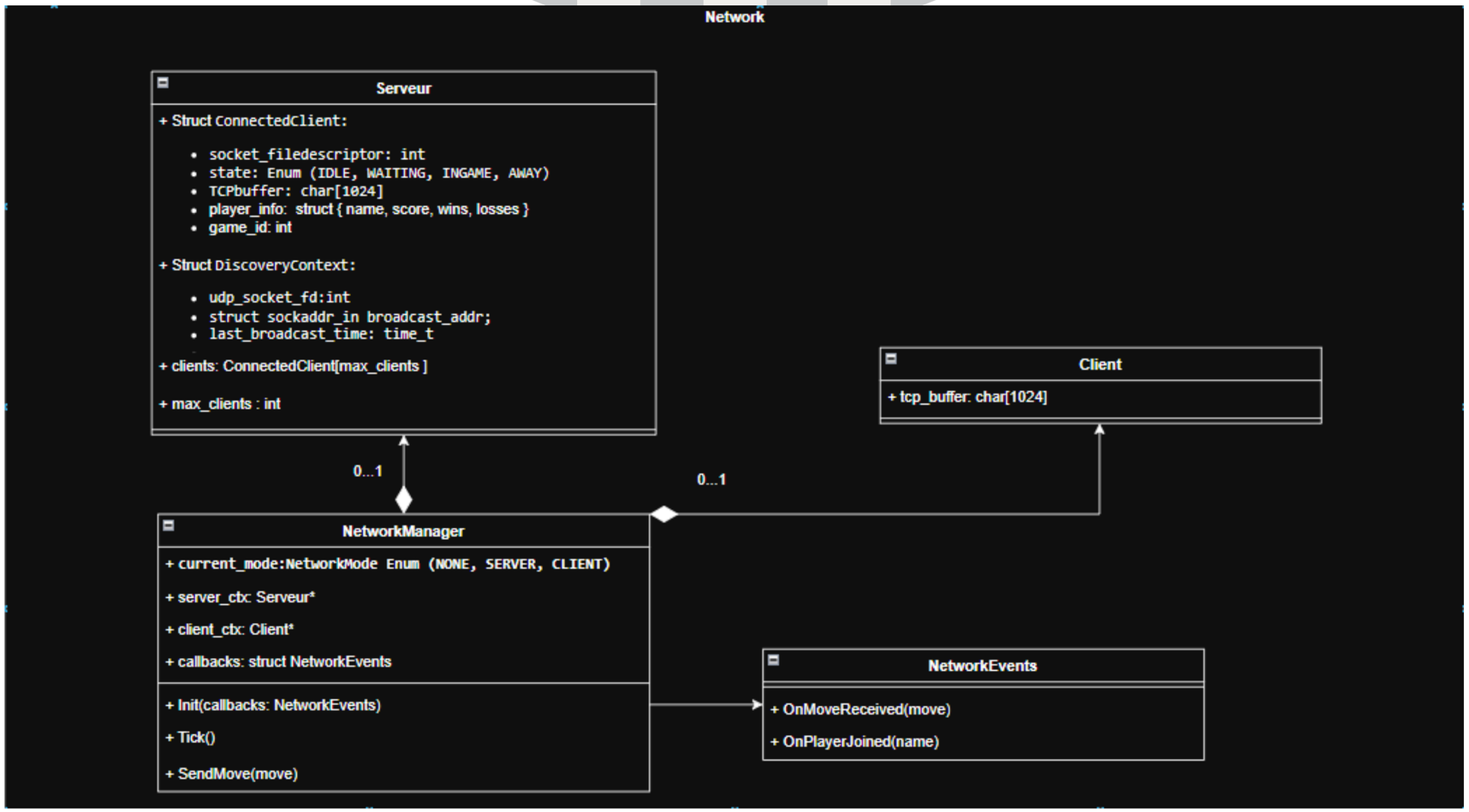
Moteur de jeu



Moteur de jeu

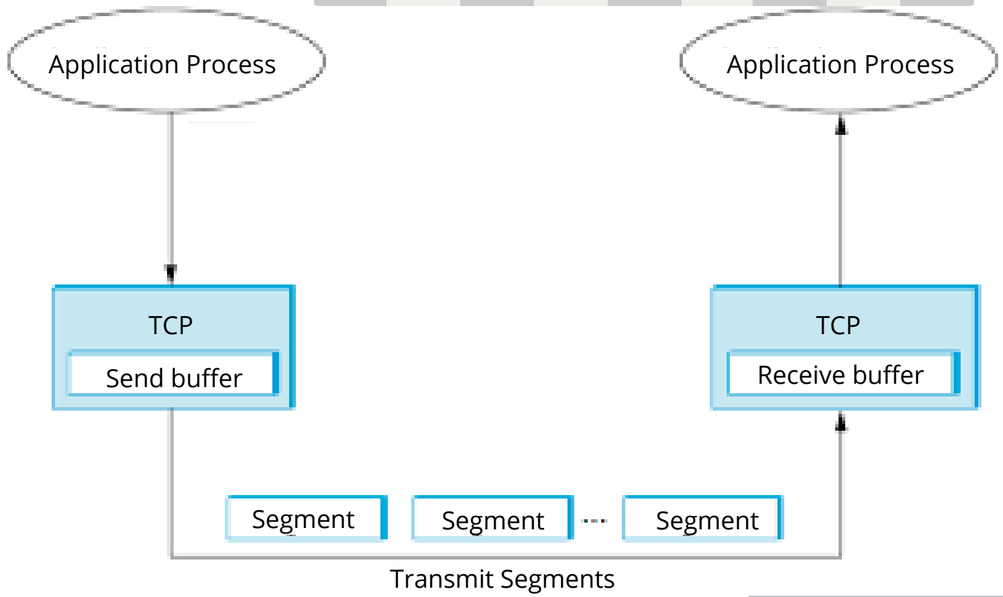


Réseau


Network
Serveur
+ Struct ConnectedClient:
socket_filedescriptor: int
state: Enum (IDLE, WAITING, INGAME, AWAY)
TCPbuffer: char[1024]
player_info: struct { name, score, wins, losses }
game_id: int
+ Struct DiscoveryContext:
udp_socket_fd:int
struct sockaddr_in broadcast_addr;
last_broadcast_time: time_t
+ clients: ConnectedClient[max_clients]
+ max_clients : int
Client
+ tcp_buffer: char[1024]
0...1
0...1
NetworkManager
+ current_mode:NetworkMode Enum (NONE, SERVER, CLIENT)
+ server_cbc: Serveur*
+ client_cbc: Client*
+ callbacks: struct NetworkEvents
+ Init(callbacks: NetworkEvents)
+ Tick()
+ SendMove(move)
NetworkEvents
+ OnMoveReceived(move)
+ OnPlayerJoined(name)

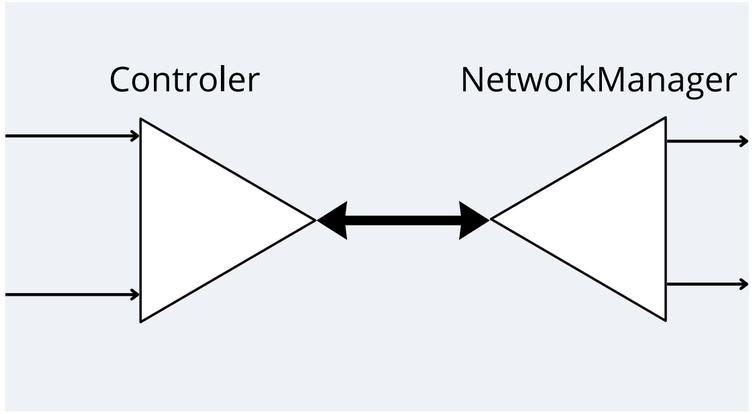
Réseau

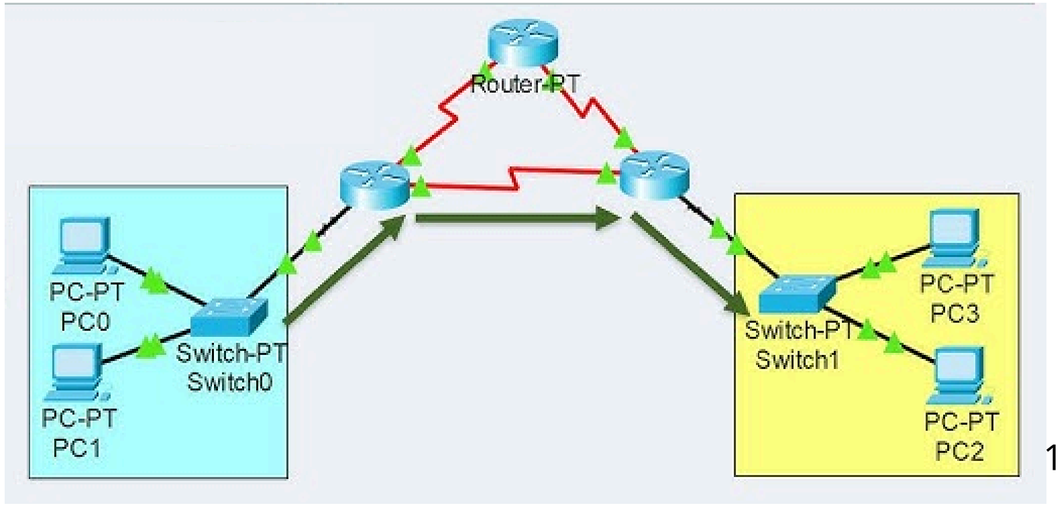
Reconstruction TCP



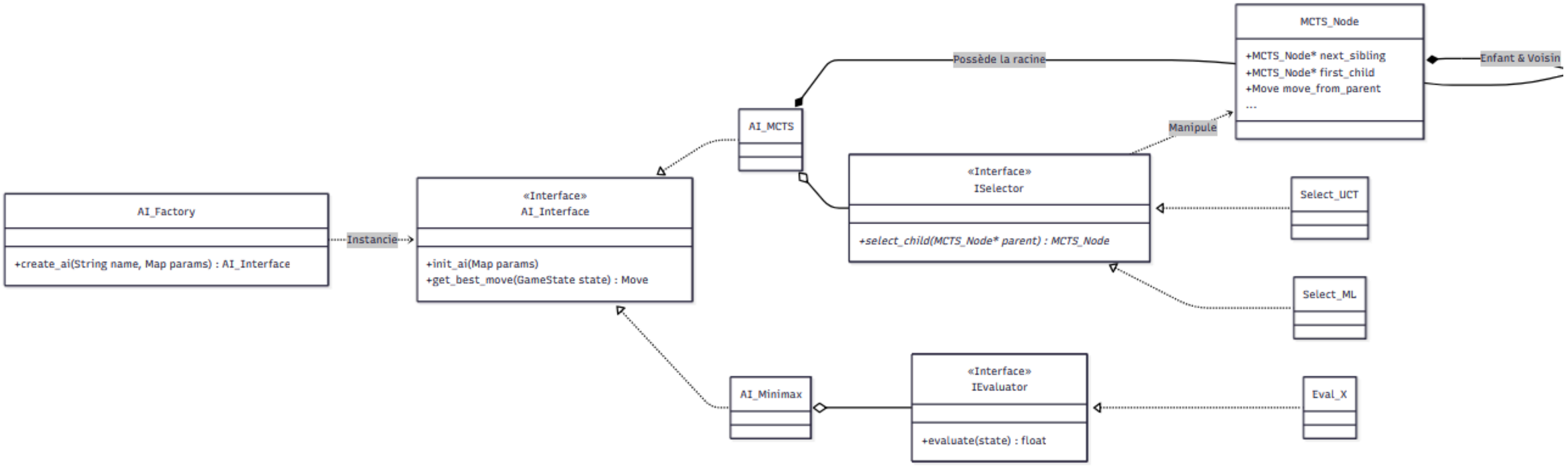
Routage Sélectif

Multiplexage (I/O)





IA


MCTS_Node
+MCTS_Node* next_sibling
+MCTS_Node* first_child
+Move move_from_parent
...
Possède la racine
Enfant & Voisin
AI_MCTS
Manipule
«Interface»
ISelector
+select_child(MCTS_Node* parent) : MCTS_Node
Select_UCT
Select_ML
AI_Factory
+create_ai(String name, Map params) : AI_Interface
Instancie
«Interface»
AI_Interface
+init_ai(Map params)
+get_best_move(GameState state) : Move
AI_Minimax
«Interface»
IEvaluator
+evaluate(state) : float
Eval_X

IA
Evaluations

Un exemple d’évaluation :
L’évaluation de Mobilité
(Territoire et Mobilité)

“Quelle est ma liberté de mouvement parmi l’espace que j’occupe ?”

IA
Selection

Sélectionner comment ?

UCT

Upper Confidence bounds applied to Trees (+ évaluation légère)

ML

Machine Learning



Gtk 4.x



Gtk 4.x

Concept de base

- GtkApplication
- GtkWindow
- GtkWidget
- GtkButton, GtkOverlay, GtkDropdown, ...
- GtkStack

Drag'n'Drop

- GtkDragSource
- GtkDropTarget

GUI

Gtk 4.x

Comment représenter les fenêtres ?

→ *Fichier XML*

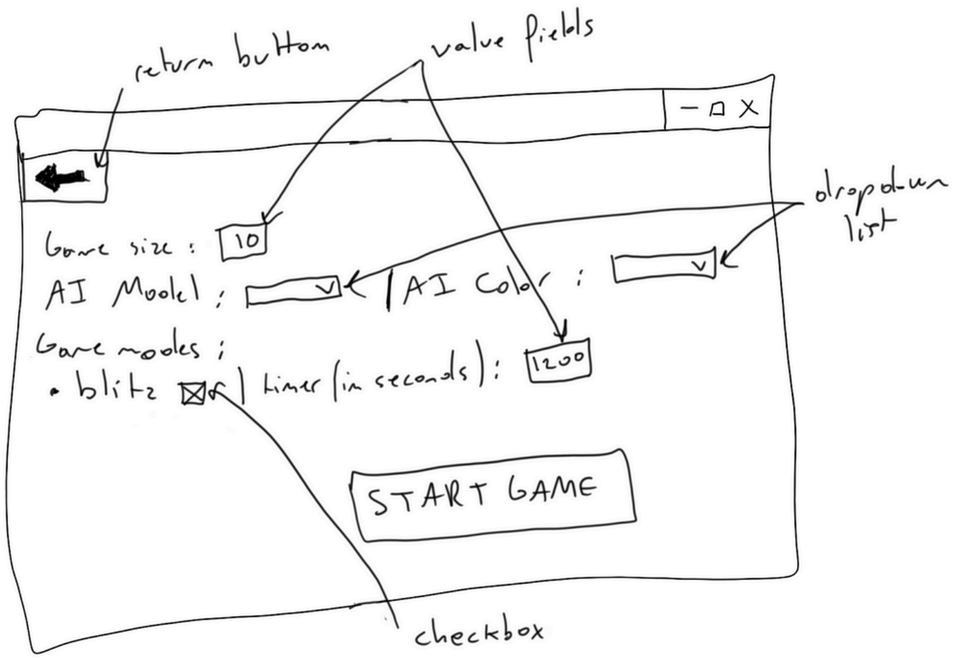
→ Simple d'utilisation

→ Fichier séparé = libère du code

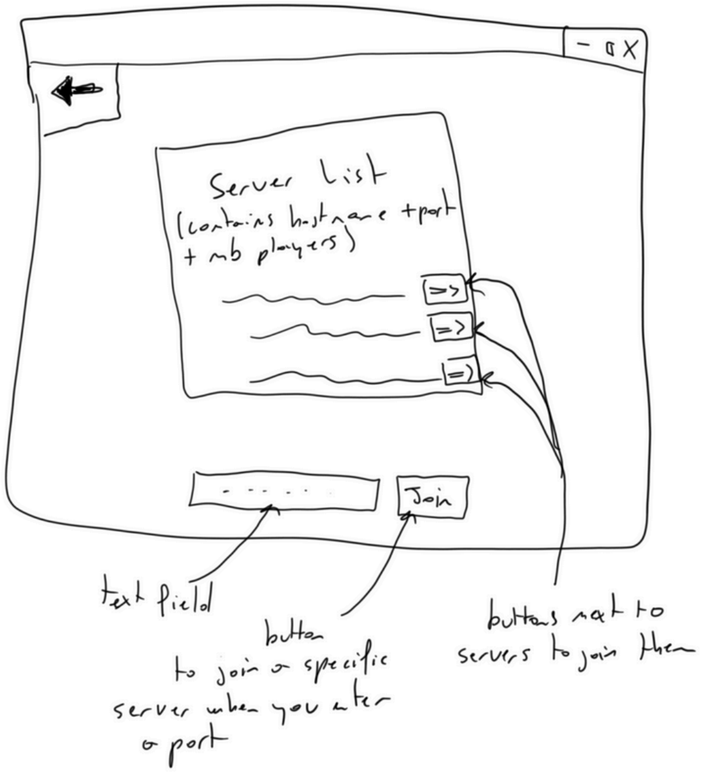
→ S'intègre facilement à Gtk

GUI

Exemples

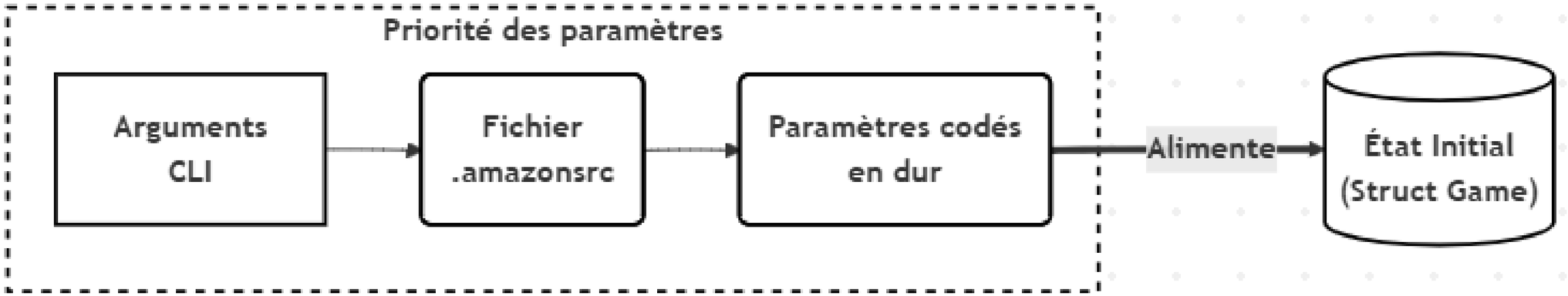


NEW GAME WINDOW
(SINGLEPLAYER)

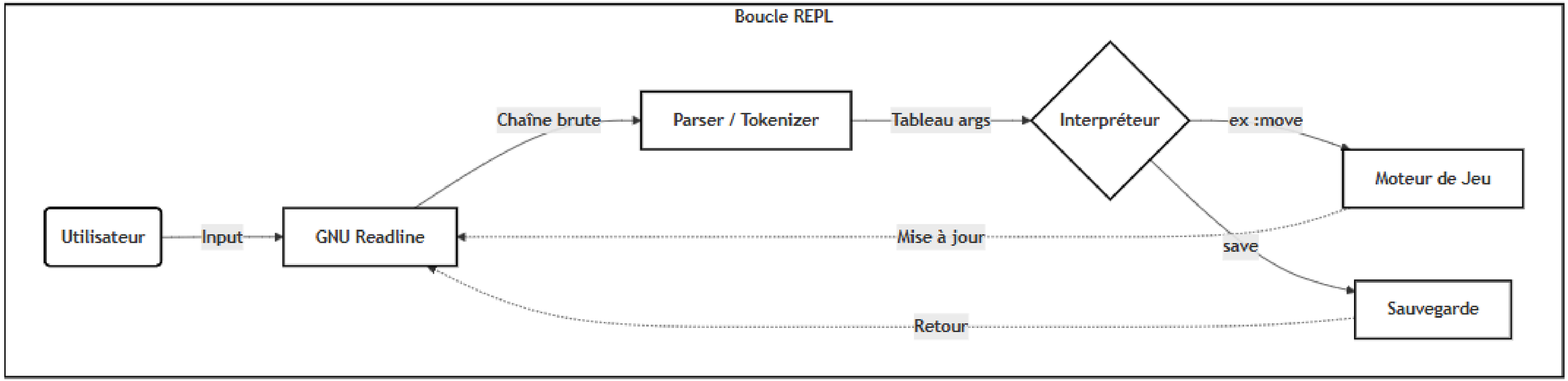


JOIN SERVER
(MULTIPLAYER)

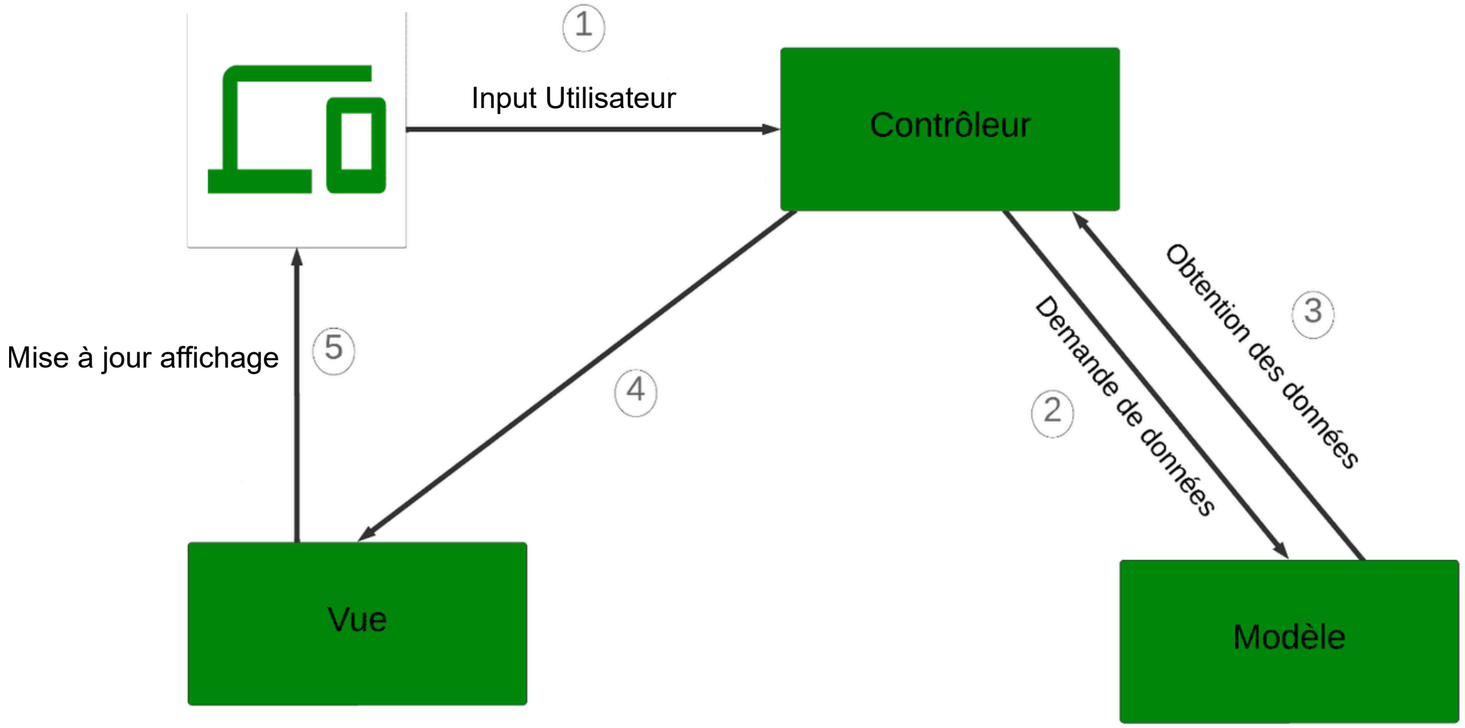
CLI


Priorité des paramètres
Arguments CLI
Fichier .amazonsrc
Paramètres codés en dur
Alimente
État Initial (Struct Game)

CLI


Boucle REPL
Utilisateur
Input
GNU Readline
Chaîne brute
Parser / Tokenizer
Tableau args
Interpréteur
ex :move
Moteur de Jeu
Mise à jour
save
Sauvegarde
Retour

Architecture



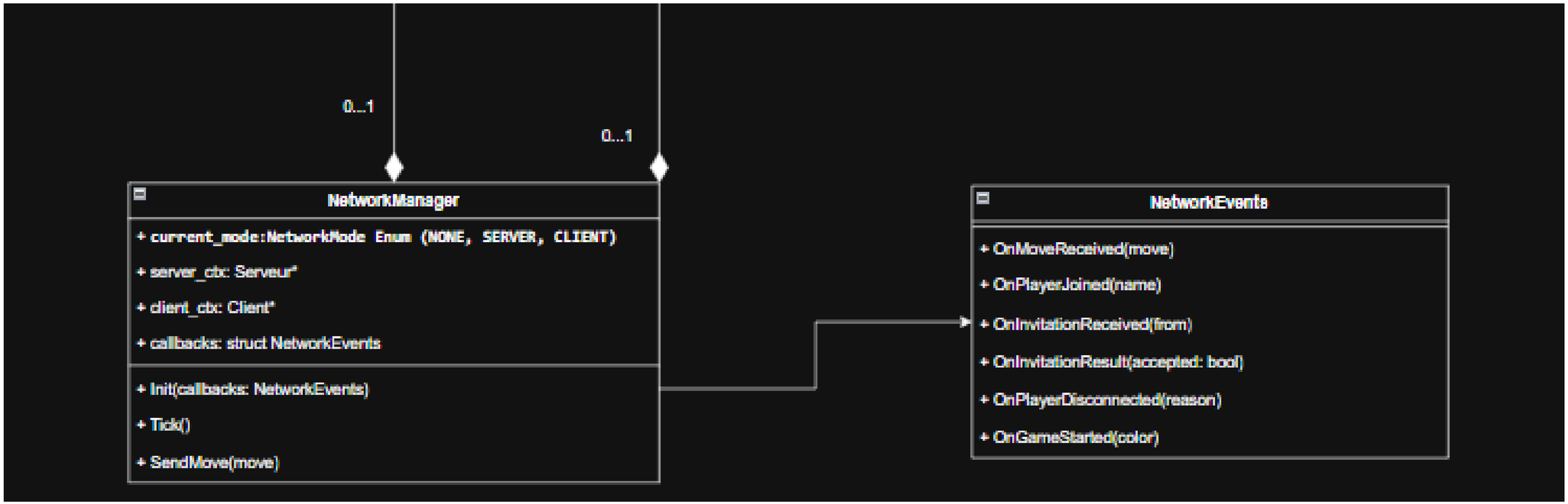
Références bibliographiques

image 1: https://cf.geekdo-images.com/lJwtJPMMJDHyh5VQEtQ0Qg__opengraph/img/28qvxq0JEzEKfKdUkJ-soUXD3w8=/fit-in/1200x630/filters:strip_icc()/pic1152391.jpg
image 2: https://cas.stthomas.edu/_media-library/_images/history/1920x1080/stthomas-history-time.jpg
image 3 : https://media.licdn.com/dms/image/v2/D4E12AQFz43ASmCZy2Q/article-cover_image-shrink_600_2000/article-cover_image-shrink_600_2000/0/1662135421412?e=2147483647&v=beta&t=3aaBzGQbZrcjTwjk57uClMbs_FlJpKuIIeBBx7uF90Q
image 4 : https://bodogemu.com/24/amazons-start-board.png
image 5 :https://bodogemu.com/24/amazons-running-game-256.png
image 6: https://static.alphorm.com/blog-alphorm-uploads/2025/01/03193647/schema-architecture-mvc-laravel-78.png
image 7 : https://preview.redd.it/hexagonal-vs-clean-vs-onion-architecture-which-is-truly-the-v0-75onmecznf0g1.png?width=481&format=png&auto=webp&s=dbcaae9872dfd77562e5e651c6a09def2e6b5a98
image 8 :https://preview.redd.it/hexagonal-vs-clean-vs-onion-architecture-which-is-truly-the-v0-b9sssngznf0g1.png?width=1050&format=png&auto=webp&s=9ec963b4b0bd9949c01006d2779eb326dfc5756b
Tree example image : https://www.researchgate.net/publication/380934806_Code_Repair_with_LLMs_gives_an_Exploration-Exploitation_Tradeoff
Routage selectif exemple image :https://www.youtube.com/watch?v=9lYJMiZ2EYA

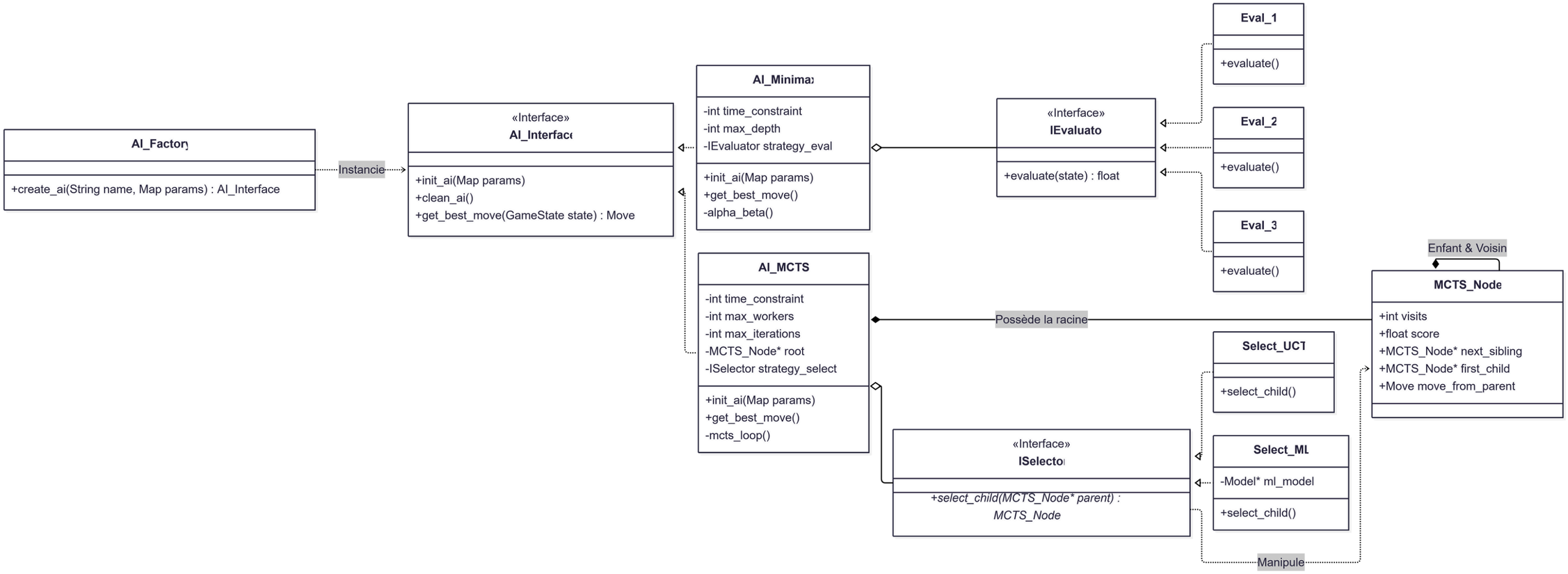
Références d'informations :
[1] Louis Victor Allis. Searching for Solutions in Games and Artificial Intelligence. Ph.D. Thesis. 1994. url : http://fragrieu.free.fr/SearchingForSolutions.pdf (visité le 21/01/2026).
[2] Patrick Hensgens. A Knowledge-based Approach of the Game of Amazons. Master's Thesis. 2001. url : https://project.dke.maastrichtuniversity.nl/games/files/msc/Hensgens_thesis.pdf (visité le 20/01/2026).
[3] John Tromp et Gunnar Farnebäck. Combinatorics of Go. 2007. url : https://tromp.github.io/go/gostate.pdf (visité le 21/01/2026).
[4] GeeksforGeeks. Left Child Right Sibling Representation of Tree. 2024. url : https://www.geeksforgeeks.org/left- child- right- sibling-representation-tree/ (visité le 21/01/2026).

Annexe


Network
Serveur
+ Struct ConnectedClient:
socket_filedescriptor: int
state: Enum (IDLE, WAITING, INGAME, AWAY)
TCPbuffer: char[1024]
player_info: struct { name, score, wins, losses }
game_id: int
+ Struct DiscoveryContext:
udp_socket_fd:int
struct sockaddr_in broadcast_addr;
last_broadcast_time: time_t
+ clients: ConnectedClient[max_clients]
+ max_clients : int
+ listening_socket : int
+ ServerStart(port)
+ ServerStop()
+ serveurStatus()
+ players(): list players
+ HandleInvite(targetID)
+ RoutePacketToOpponent(packet, client_source)
+ scoreboard():struct scoreboard
+ Player(player_ID): struct player
+ newPlayerID()
Client
+ discovered_servers: List<ServerInfo>
+ tcp_buffer: char[1024]
+ tcp_socket: int
+ udp_socket: int
+ last_message_time: time_t
+ Join(ip*,port)
+ quit()
+ ping()
+ accept()
+ decline()
+ cancel()
+ away()
+ back()
+ GetServerList(): List
+ ListenForBroadcasts()
+ SendPacket(type, payload)


0...1
0...1
NetworkManager
+ current_mode:NetworkMode Enum (NONE, SERVER, CLIENT)
+ server_cbx: Serveur*
+ client_cbx: Client*
+ callbacks: struct NetworkEvents
+ Init(callbacks: NetworkEvents)
+ Tick()
+ SendMove(move)
NetworkEvents
+ OnMoveReceived(move)
+ OnPlayerJoined(name)
+ OnInvitationReceived(from)
+ OnInvitationResult(accepted: bool)
+ OnPlayerDisconnected(reason)
+ OnGameStarted(color)

IA


AI_Factory
+create_ai(String name, Map params) : AI_Interface
Instancie
«Interface»
AI_Interface
+init_ai(Map params)
+clean_ai()
+get_best_move(GameState state) : Move
AI_Minimax
-int time_constraint
-int max_depth
-IEvaluator strategy_eval
+init_ai(Map params)
+get_best_move()
-alpha_beta()
«Interface»
IEvaluato
+evaluate(state) : float
Eval_1
+evaluate()
Eval_2
+evaluate()
Eval_3
+evaluate()
AI_MCTS
-int time_constraint
-int max_workers
-int max_iterations
-MCTS_Node* root
-ISelector strategy_select
+init_ai(Map params)
+get_best_move()
-mcts_loop()
Possède la racine
Enfant & Voisin
MCTS_Node
+int visits
+float score
+MCTS_Node* next_sibling
+MCTS_Node* first_child
+Move move_from_parent
«Interface»
ISelecto
+select_child(MCTS_Node* parent) : MCTS_Node
Select_UCT
+select_child()
Select_ML
-Model* ml_model
+select_child()
Manipule

IA
Evaluations

Déjà… évaluer quoi ?

Le “potentiel” d’un coup
La liberté d’action d’une amazone
Un rapport de force sur les cases

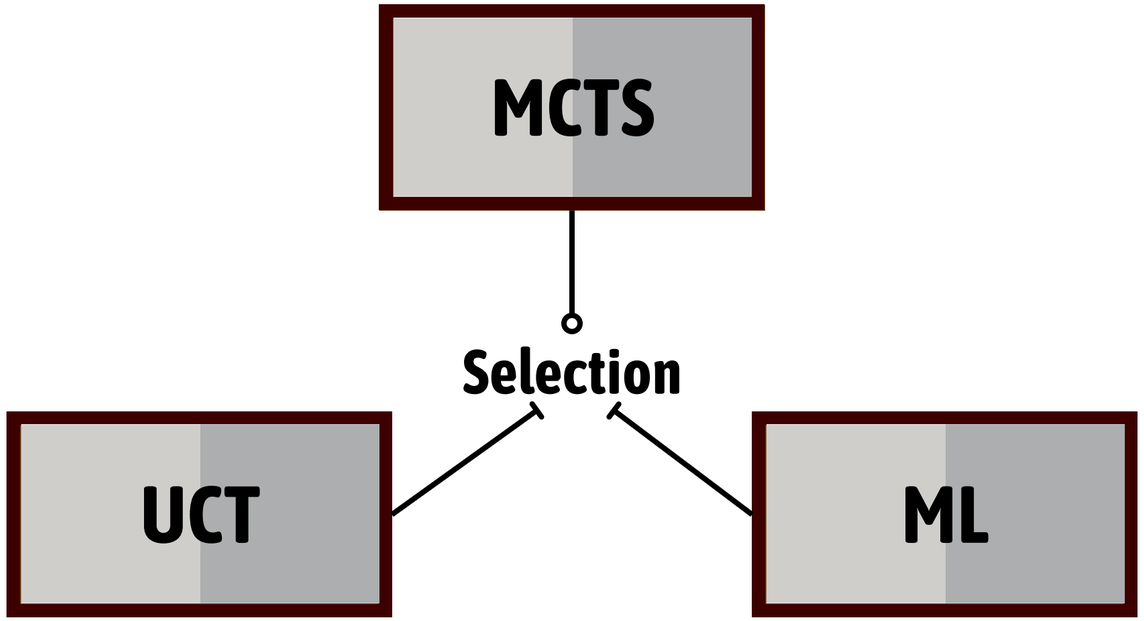
IA

Evaluations

TaM	QD&KG	ME
remplacé par TeM		

IA

Selections



GUI

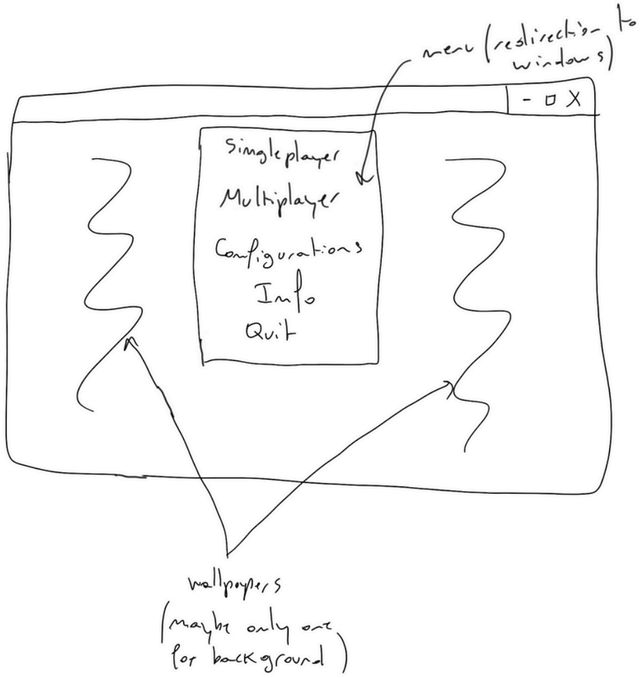
Gtk 4.x

Exemple de structure

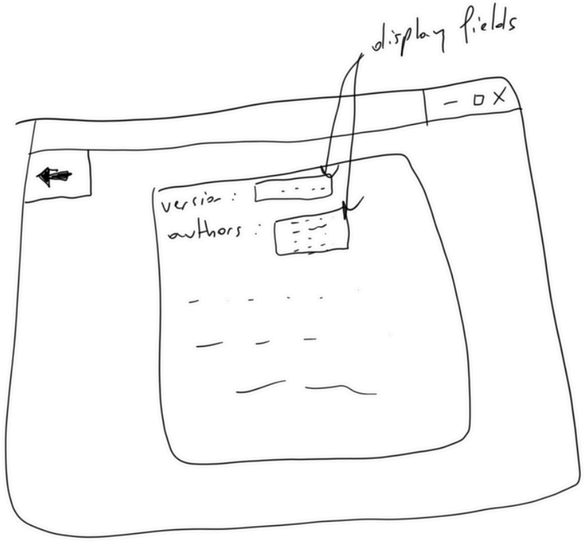
- GtkApplication
  - GtkWindow
    - GtkOverlay
      - GtkStack
        - GtkWidget (Menu principal)
          - GtkButton
          - ...
        - GtkWidget (Menu singleplayer)
        - ...

GUI

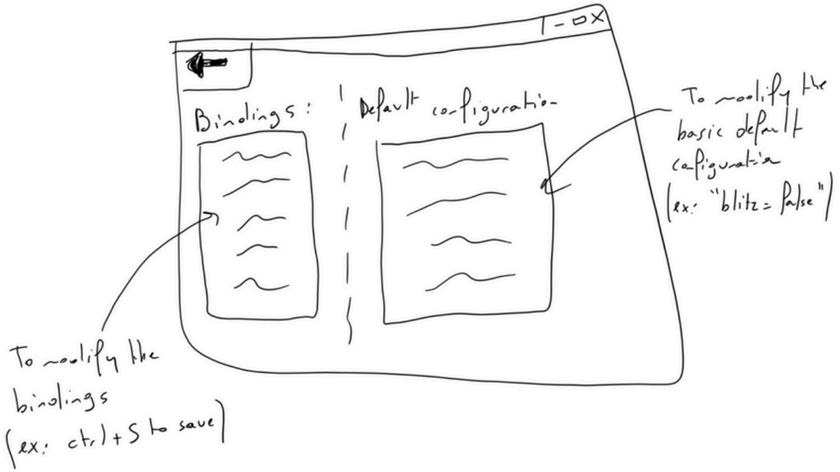
Autres schémas



MAIN MENU



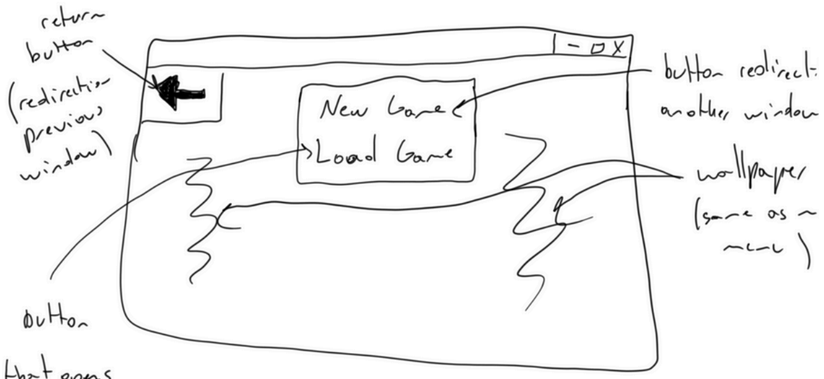
INFO WINDOW



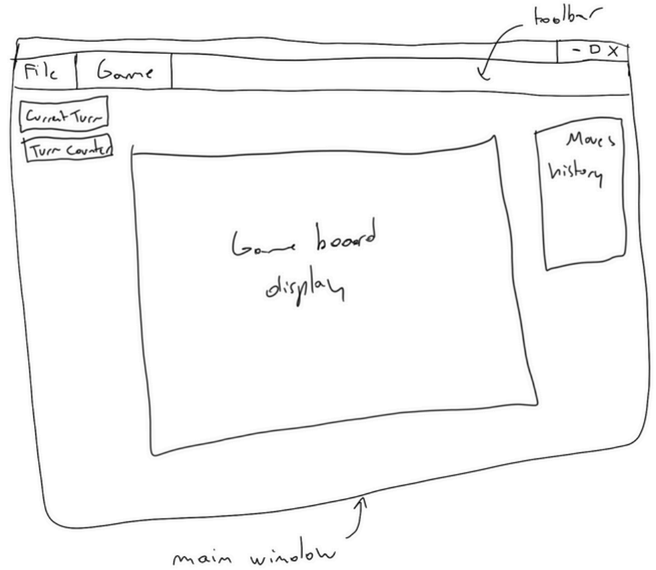
CONFIGURATIONS PANEL

GUI

Autres schémas



SINGLE PLAYER WINDOW



When in a game and press Quit without saving before (in "File" tab or the upright cross), it shows a pop-up asking if we want to save before or not

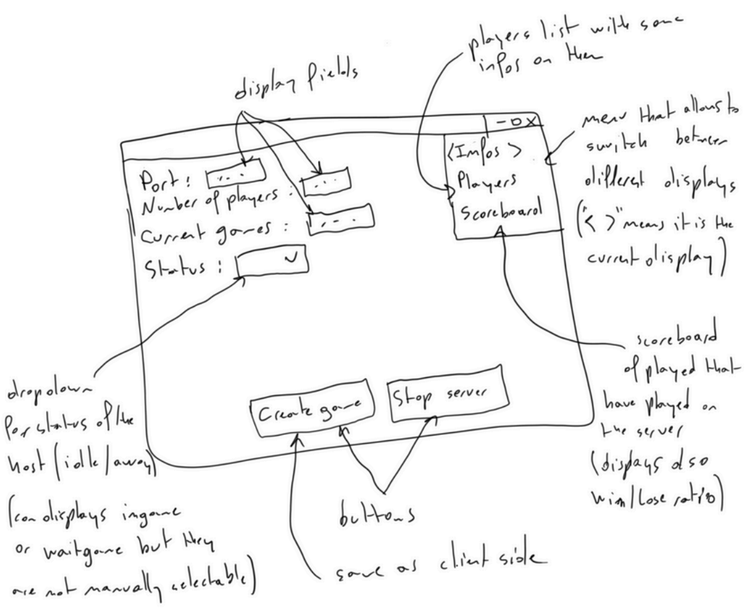
The players get a popup when a game is finished, telling them if they won or lost

⚠ Undo/Redo disabled in multiplayer, and load/save only available for the "game master"

GAME WINDOW

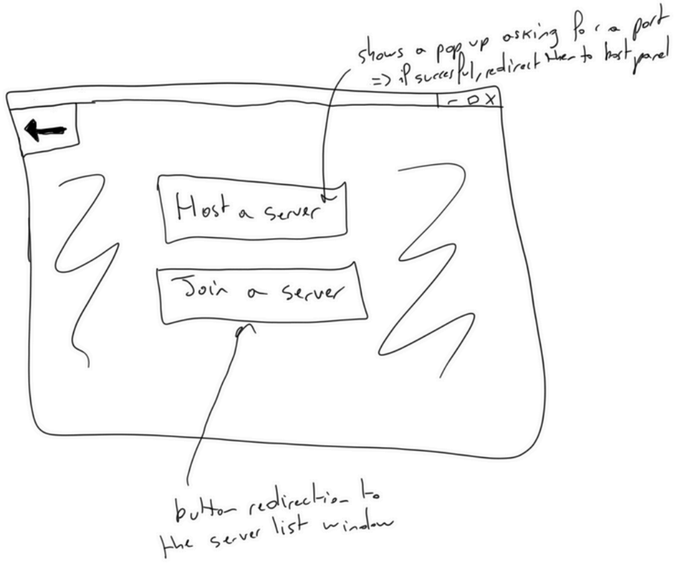
GUI

Autres schémas



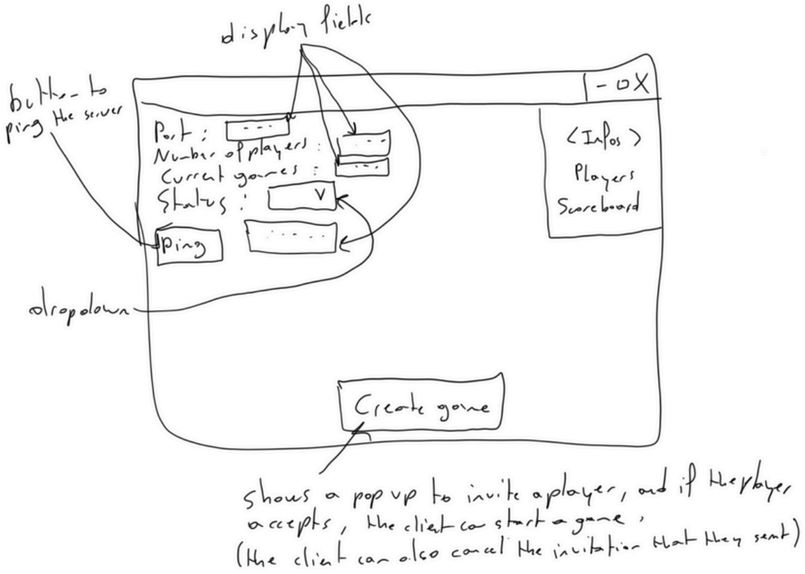
HOST PANEL WINDOW

(MULTIPLAYER)



MULTIPLAYER

WINDOW



(but you can also just decide to use one or two AI instead) (and you can set some settings like blitzcrit)
the client gets a popup when they receive or pop to accept or refuse the invitation

CLIENT PANEL WINDOW

(MULTIPLAYER)